

Master thesis proposal on topic:

Dynamic social and symbolic cues in the context of social anxiety and depressive personality trait

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Abstract

Our environment is crowded with objects that often require further scrutiny, and attending to task-relevant features of the environment is essential for effective functioning. However, attention is a finite resource and must be selectively deployed. Consequently, people use various directional cues to focus their attention on important events and objects in the world. These cues can be symbolic, such as pointing arrows or words with directional meaning. Additionally, as highly social creatures, humans also rely on social cues like gaze direction, head orientation, or pointing gestures. The purpose of this study is to determine whether attentional focus is modulated by anxious or depressive personality traits in the context of social and symbolic cueing. This insight will enhance our understanding of how people experience and navigate their social world. To achieve this, we will conduct a study using a variation of the gaze cueing task, where the cueing stimuli (a gazing face for social cueing or an arrow for symbolic cueing) presented centrally are entirely irrelevant and can therefore be easily ignored by the participants. Additionally, we will utilize the Beck Depression Inventory - II (BDI-II) (Beck et al., 1961) and The Social Phobia Inventory (SoPhI, Moore & Gee, 2003) to measure individual anxiety and depression levels as well as other personal traits.

Introduction

Y

Symbolic vs. social cues

The study of gaze cueing, particularly in comparison with arrow cues, unveils intricate dynamics between perceptual and attentional systems in processing social and symbolic cues. This extensive area of research, was previously focused on the investigation of gaze-cueing (GC) automaticity, now is significantly expanded and more oriented towards comparing attentional shifts triggered by symbolic and social cues and their modulation.

For instance, Hermens (2018) discovered that gaze cueing had a weaker effect than arrows and hands in both adults and children when the cue is not at the fixation point. Burton et al. (2009) observed that rotated heads and pointing hands interfered with responses to gazing eyes, but not vice versa. This is corroborated by a neuroimaging study by Hietanen et al. (2006), which revealed that gaze- and arrow-cued attentional orienting are subserved by distinct

neural networks, with arrow cues more actively engaging the voluntary attention network.

Another line of neuroimaging research, particularly in participants with disconnected hemispheres, demonstrated differences in information processing. A study by Kingstone et al. (2000) found that attention shifts towards eye gaze primarily involve activation in the right hemisphere (in patients with hemispheric disconnection) specialized for face processing, in contrast to arrows, which trigger bilateral activation, highlighting the unique nature of gaze cues in cognitive processing). The results were replicated in several other studies (Ristic et al., 2002; Greene & Zaidel, 2011).

Recent studies have explored these dynamics further, finding that the Gaze Cueing Effect (GCE) is maximized when group members uniformly direct their gaze (Sun et al., 2017). The introduction of inconsistent cues, such as faces looking in opposite directions, reduces the GCE, suggesting that individuals tend to follow a “majority rule” when exposed to multiple cueing (Sun et al., 2017). In addition, Capozzi et al. (2018) observed a “quorum-like” order in small groups, where the need for consistent facial cues

to elicit an attentional response increases with group size.

Another notable difference between the type of cueing lies in how people select objects in response to these cues, with eye gaze eliciting a more specific attentional orientation than arrows (Marotta et al., 2012). Marotta et al. (2012) hypothesized that biologically and socially relevant gaze cues encourage more specific attentional orienting compared to arrow cues. Consistent with this hypothesis, they found that eye-gaze cues focus attention precisely on a specific location or part of an object, in contrast to arrow cues which distribute attention over the entire object. More recent research examined how attention is allocated when placeholders are **either** grouped based on Gestalt's law of proximity. This study revealed that arrows disperse attention across a group of objects, whereas eye-gaze narrows attention to a specific target. This effect was absent when placeholders were ungrouped or not present (Chacón-Candia et al., 2023).

In summary, these findings suggest that the attention-shifting processes elicited by these types of cues are distinct, both neurobiologically and mechanistically. Comparing them provides insights into our cognitive understanding of symbols and social cues, highlighting the importance of our social nature and its varying significance across individuals.

Social cueing (gaze) modulation

Although the distinction between gaze and arrow cues in guiding attention is well-recognized, the specific factors that modulate the GCE remain less certain. Despite the topic's popularity and extensive research, findings have been inconsistent and often lack **replicability**. This ambiguity underscores the need for further investigation to clarify the factors influencing GCE.

The GCE is influenced by a complex array of factors that include individual psychological characteristics, situational variables, and broader

social and cognitive contexts. These factors can be broadly categorized into several key groups that impact the GCE: (a) characteristics of the observer, (b) characteristics of cueing faces, and (c) their relationship (Dalmaso et al., 2020). Each category encompasses a range of elements that collectively shape the nuanced ways in which gaze cues are processed and acted upon. This complexity contributes significantly to the mixed data in the field, highlighting the challenge of isolating and identifying the precise mechanisms at play in gaze cueing, which leads to the mixed and sometimes contradictory results observed across different studies.

An extended review provided by Dalmaso et al. (2020) brings more clarity and organization into the field.

Evidence suggests that **higher dominance, social status, and potentially trustworthiness can** elicit a greater orienting response, reflecting the social relevance associated with these attributes (Koski, Xie, & Olson, 2015, as cited in Dalmasso, 2020). Physical dominance, often linked to hierarchy and defined as the use of force and intimidation to influence others (Henrich & Gil-White, 2001, as cited in Dalmasso, 2020), is commonly associated with masculinity. Masculine faces are typically perceived as more dominant than feminine faces (Perrett et al., 1998, as cited in Dalmasso, 2020). Studies using masculinized and feminized faces have shown that masculinized faces elicit a stronger GCE, especially at shorter SOAs (Jones et al., 2010, as cited in Dalmasso, 2020). Further research confirmed that **dominant-looking male faces induced a stronger GCE compared to nondominant female faces**. Moreover, nondominant female faces only induced a reliable GCE when paired with nonthreatening images, suggesting that in threatening contexts, physically weaker individuals are perceived as less capable of offering safety (Ohlsen et al., 2013, as cited in Dalmasso, 2020). **Social status, often derived from intellectual abilities and respect from others** (Gould, 2002), also influences the GCE. Dalmaso et al. (2012) found that faces associated with higher social status elicited a greater GCE. This effect, as further shown by Dalmaso et al. (2014), tends to diminish over time; with a

longer SOA, the magnitude of the GCE was not affected by the social status of the gaze cue provider, suggesting that the influence of social status on GCE is reflexive and occurs early (Dalmasso, 2014). Regarding trustworthiness, the results are less clear. While Süßenbach and Schönbrodt (2014) reported a greater GCE in response to faces deemed trustworthy, other studies have found no such modulation (Bayliss & Tipper, 2006; King, Rowe, & Leonards, 2011; Strachan, Kirkham, Manssuer, Over, & Tipper 2017, as cited in Dalmasso, 2020)."

Facial expressions, combined with gaze direction, play a crucial role in social communication by indicating the presence of relevant objects or threats. However, studies examining the GCE in response to facial expressions, particularly static ones, have yielded mixed results. Several studies (Hietanen & Leppänen, 2003; Bayliss et al., 2007; Holmes et al., 2010; Slessor et al., 2008) found no modulation of the GCE by emotional expressions.

Conversely, some studies have identified modulatory effects under specific conditions. For instance, a greater GCE for happy faces, particularly from female actors or in low AQ individuals (Hori et al., 2005; Hudson et al., 2012 as cited in Dalmasso, 2020) or an increased GCE in response to angry, happy, and fearful faces under high cognitive load or after oxytocin administration (Pecchinenda and Petrucci, 2016; Tollenaar et al., 2013, as cited in Dalmasso, 2020) were reported. Dynamic stimuli like morphing faces have also shown greater GCE responses to fearful or angry faces (Bayless et al., 2011; McCrackin & Itier, 2018), although not consistently (Fichtenholtz et al., 2007, 2009, as cited in Dalmasso, 2020).

Individual differences of the perceiver also significantly influence the GCE. Mathews et al. (2003) found a greater GCE in response to fearful faces in high-anxious individuals, a pattern not consistently replicated (Galfano et al., 2011; Holmes et al., 2010; McCrackin & Itier, 2019, as cited in Dalmasso, 2020). Other studies reported variations in the GCE related to fear, disgust, anger, and high AQ scores (Pletti et al.,

2015; McCrackin & Itier, 2019). The salience of affective information and frequency of exposure to emotional expressions also shape the GCE (Bayliss et al., 2010; Friesen et al., 2011; Kuhn et al., 2016a, as cited in Dalmasso, 2020). Overt attention paradigms, such as instructed saccade paradigms, have yielded mixed results (Bonifacci et al., 2008; Matsunaka & Hiraki, 2019, as cited in Dalmasso, 2020).

Moreover, an individual's personality traits can shape their social attention abilities. Bayliss, di Pellegrino, and Tipper (2005) found that the GCE was stronger in females across all SOAs, with a more pronounced effect at the longest SOA. They also discovered that using nonpredictive arrows, instead of gaze cues, revealed a significant gender difference in attention orientation, suggesting that males and females process symbolic cues differently. This finding supports Baron-Cohen's (2002) "extreme male brain" theory of autism, which posits that males may exhibit more autistic-like traits and reduced social abilities compared to females, influencing the Gaze Cueing Effect (GCE).

Individuals with lower self-esteem have been linked to higher GCE (Wu et al., 2014), likely due to their increased need for social reconnection (e.g., Baumeister & Leary, 1995 as cited in Dalmasso, 2020). Familiar faces, such as those of colleagues from the same department, elicited a stronger GCE compared to faces of individuals from different departments, particularly in female participants, indicating heightened sensitivity to eye-gaze cues (Bayliss et al., 2005).

Introversiion/extroversion traits, combined with emotional expression, also influence the GCE. Introverts showed a reliable GCE in response to neutral and happy faces but not to angry faces, whereas extroverts demonstrated the opposite pattern (Ponari et al., 2013).

The GCE is affected not only by stable personality traits but also by temporary internal states. Feelings of social acceptance or rejection prior to the GCE task can modulate the effect (Wilkowski et al., 2009 as cited in Dalmasso, 2020). Similarly, Cui et al. (2014) found that

high or low power priming altered gaze cueing responses, suggesting the GCE's sensitivity to perceived social hierarchy changes. Contrarily, Capellini, et al. (2019) explored the effect of a temporarily induced state of belongingness on the GCE using a gaze-cueing task and the cyberball task, finding that this manipulation influenced the GCE in a direction opposite to that reported by Wilkowski et al. (2009, as cited in Dalmasso).

a conclusion for this section is needed.

Anxiety and Gaze cueing

The inconsistency in research continues in studies investigating the effect of anxiety traits or social anxiety phobia on gaze cueing (GC).

A recent study by Talipski et al. (2020) investigated whether social anxiety influences how people respond to others' gaze directions, especially when combined with different facial expressions. The hypothesis was that social anxiety **might modulate gaze-cueing effects, particularly for socially threatening expressions like anger.** A sample of 100 female participants was studied using a gaze-cueing task involving images of faces with angry, fearful, and neutral expressions. The results showed large and reliable gaze-cueing effects, but these were not modulated by social anxiety or other individual difference variables like trait anxiety, depression, and autistic-like traits. This outcome suggests that while gaze direction is a significant social cue, its influence is not significantly altered by social anxiety levels. This result aligns with a study by Gregory et al. (2019), where a similar lack of modulation by social anxiety was observed in a study of 105 participants divided into high and low social anxiety groups. Although high social anxiety participants spent more time looking at actors' faces initially, there were no significant differences in gaze-following behavior or first fixation metrics between the groups. Supporting these findings, several other studies (Lassalle & Itier, 2015b; McCrackin & Itier, 2019; Neath et al., 2013; as cited in Talipski et al. 2020) also failed to consistently observe anxiety-related modulations of GC. However, the replicability of this finding has

been **questioned**, as opposite results have also been reported.

An earlier study by Holmes et al. (2006) employed a design where emotional and neutral faces were presented in peripheral locations, either to the left or right of the fixation point. **Contrary**, it was found that gaze-related spatial cueing effects to laterally presented fearful faces and centrally presented angry faces were significantly modulated by the anxiety levels of the participants. Specifically, individuals with high levels of state anxiety exhibited **enhanced shifts of attention** compared to those with low state anxiety.

Furthermore, higher levels of trait anxiety have been associated with a greater GCE for fearful gaze cues (Putman et al., 2006), indicating a potential hypervigilance to threat among anxious participants.

Additionally, in a 2016 study by Boll et al. researchers conducted two eye-tracking experiments with 22 participants to investigate gazing behavior towards facial cues, especially in the context of social phobia. The first experiment focused on reflexive attention shifts towards the eyes or mouth of emotional faces (angry, fearful, happy, neutral), presented both briefly and for extended durations. The goal was to examine early attentional shifts and differentiate them from sustained attention. The hypothesis suggested that individuals with social phobia would demonstrate increased reflexive attention to the eyes, with variations depending on the emotional expression of the faces. The second experiment utilized a gaze-cueing paradigm with non-predictive gaze cues, aimed at examining gaze-following behaviors and how they might assist in target identification. The researchers hypothesized that there would be notable differences in gaze-following and gaze-cueing effects between patients with social phobia and control participants, influenced by initial attentional focus and potential avoidance behaviors.

Results from the first experiment indicated no significant differences in accuracy or response latency between the two groups. However, it was

observed that, in general, participants made more initial saccades towards the eyes than the mouth. Notably, individuals with social phobia showed a more pronounced bias towards the eye region. The influence of emotional expression was evident in both groups, with more saccades directed towards the eye region for fearful and neutral expressions compared to happy faces, and more towards the mouth for happy faces (Boll et al., 2016).

In the context of gaze cueing, both groups showed similar results in the success of the categorizing task, though participants with social phobia were significantly slower, a difference attributed to longer reaction times. When analyzing faster reaction times relative to the median, a significant gaze cueing effect was observed. The eye movement data revealed that participants more frequently directed their attention towards the gaze direction of the depicted faces rather than the opposite direction. Interestingly, a group effect was noted; patients with social phobia exhibited gaze following behaviors, but these were significantly reduced for the initial saccade compared to the second, indicating an interaction between congruency and saccade type. Additionally, in the social phobia group, there was a noticeable effect of emotion, with more congruent saccades occurring for angry and fearful emotional expressions (Boll et al., 2016).

thus, overall, research suggests that

Depression and Gaze cueing

Less data has been reported regarding the relationship between depression and gaze cueing (GC).

In a comprehensive study that combined data from seven experiments, researchers explored how emotional modulation of gaze-cuing varies in the general population, taking into account factors such as participant sex and self-reported subclinical traits like anxiety, depression, and autistic traits (McCrackin et al., 2018). Their findings revealed several key insights: first, emotional enhancement of gaze-cuing can occur with both positive and negative expressions. Second, participants scoring higher on the

Attention to Detail subscale of the Autism Spectrum Quotient showed a reduced emotional enhancement of gaze-cuing, especially with happy expressions. Third, the emotional modulation of gaze-cuing was not significantly influenced by participants' levels of anxiety, depression, or sex, although it was observed that women generally demonstrated a larger overall gaze-cuing effect compared to men (McCrackin and Itier, 2018).

Similar results concerning personality traits come from another study that examined patients with major depressive disorder. In this study, 21 patients and 21 healthy age-matched controls participated in an object categorization task (Li et al., 2012). The task involved a schematic face gazing either towards or away from the objects that participants had to categorize, followed by rating their preference for these objects.

The findings indicated that both the patient group and the control group categorized objects faster when gazed at compared to when not gazed at, implying that patients with depression did not exhibit impairments in their ability to respond to gaze cues. However, a noticeable difference was observed in preferences: while healthy controls showed a marked preference for objects that were gazed at, patients with depression did not demonstrate a significant preference for either gazed or non-gazed objects. This suggests that the primary deficit in patients with depression lies in aspects of social cognition rather than basic attentional processes (Li et al., 2012).

The current understanding of gaze cueing and its modulation by psychological traits like anxiety and depression presents a complex and often contradictory landscape. To address these inconsistencies, our proposed study will employ a novel gaze cueing paradigm with a simplified context, using neutral faces and arrows. We aim to examine the effects of anxiety and depressive traits, hypothesizing that higher rates of depression will correlate with a weaker gaze cueing effect, while higher anxiety rates will correlate with a stronger effect. By using a simpler experimental setup and comparing these findings with control trials using arrows, we anticipate gaining clearer insights into how these

psychological traits influence attentional mechanisms.

Method

The proposed study will utilize a reaction time-based methodology to measure the Gaze Cueing Effect (GCE) and the attention shift effect triggered by arrows – Arrow Cueing Effect (ACE). In our study, we are slightly diverging from the traditional central cueing paradigm, which typically displays the cueing image at the screen's center and the targets in the left or right area. Instead, we are implementing a different procedure in which we present the cue at the top center of the screen (aligned at 25% of the vertical axis) with the target appearing below it in each trial. This approach allows participants to focus exclusively on the target area, potentially ignoring the cue direction. This approach was tested in a previous experiment, and we were able to observe a significant difference between the congruent (cueing down) and incongruent (cueing up) trials. Reaction times for the categorization task will be measured. To assess anxiety and depression levels, widely used questionnaires will be employed: The Social Phobia Inventory (SoPhI, Moore & Gee, 2003) and the Beck Depression Inventory - II (BDI-II) (Beck et al., 1961).

Design

Two experiments will be conducted: the first will be mixed (cueing stimulus displayed in one randomized chunk), and the second will be blocked (the stimulus displayed sequentially in two counterbalanced blocks). This approach allows us to observe if the constant change of the cueing stimuli, requiring higher levels of cognitive load, will affect the results. The experimental setup and trial procedure are similar for both experiments, employing a symbolic cue (animated arrow) and a social cue (face with animated gaze). Both cues will be valid in 50% of the trials, ensuring a balanced design. Each experimental condition will consist of 30 trials, resulting in 240 trials per participant in total. Given this large number of trials, a five-

minute break will be proposed between the two experiments. In both experiments, cues will be centrally displayed above the target area, and a categorization task will be used.

Stimuli and Trial Procedure

Two types of cueing stimuli (CS) will be used – a social cue (a neutral face with moving pupils) and a symbolic cue (a rectangle transforming into an arrow). The CS may be presented in one of two conditions - incongruent (gazing up or up-facing arrow) and congruent (gazing down and down arrow). The target stimuli will be either an "F" or "L" letter. The categorization task will require participants to press the corresponding button when a given letter appears. In each trial, the CS is displayed for 350ms, positioned at the top center of the screen (25% off the vertical axis). The animation comprises six images (per condition) showing movement, each lasting 30ms, except the final image, which lasts 200ms. After the animation, the CS remains while the target (an "F" or "L") appears below at the bottom center (70% vertical alignment). Participants respond by pressing the chosen button, based on the target. The target stays on the screen for 1500ms or until a response button is pressed. Feedback is provided during practice trials. Each trial concludes with a 1-second blank inter-trial interval (ITI).

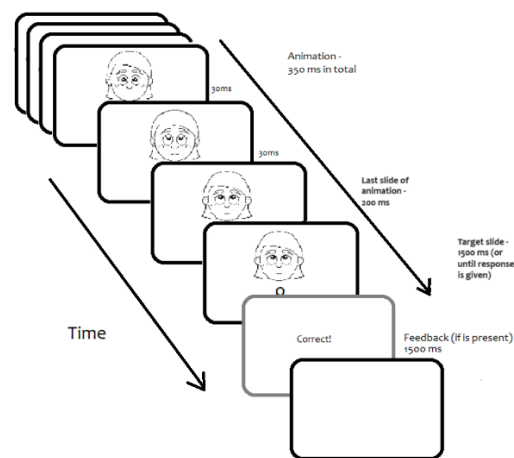


Figure 1.

Schematic representation of the time course of a trial.

Participants

Adult participants aged 18-50 will be recruited, with varied levels of anxiety and depression assessed via standardized questionnaires. The wide age range ensures broad representation of the adult population. Individuals with neurological disorders or vision impairments that might interfere with eye-tracking will be excluded.

Analysis

Data will be analyzed using ANOVA to compare reaction times across different conditions and participant groups. Significant differences in reaction times between the face cue and arrow cue conditions, in relation to participants' anxiety and depression levels, will be used to evaluate the hypotheses.

Significance and Conclusion

While symbolic cues require cognitive processing to interpret their meaning, gaze cues leverage our inherent social cognitive abilities. The human brain is particularly attuned to the gaze direction of others, interpreting it as a significant social signal. This sensitivity to gaze direction starts from infancy, indicating its deep-rooted place in our social development.

This study aims to provide a clearer understanding of how anxiety and depression may influence attentional processing in gaze cueing. By employing dynamic cues and a balanced experimental design, the study seeks to contribute to the existing literature by clarifying the relationship between psychological traits and attentional mechanisms. Our findings could have implications for understanding attentional processes in these populations, potentially adding more clarity to the field and stimulating further research. The study's conclusion will discuss the findings in the context of existing research, their practical implications, and suggest avenues for future research.

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